Geometrically Inductive Self-Supervised Learning for EEG

A Design Study of Spatial Attention Across Sessions, Subjects, and Montages

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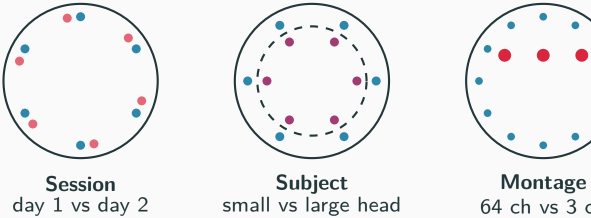
Problem: EEG is non-stationary – and the geometry is what changes

EEG drifts on **three geometric axes**:

- **Session** — cap shifts day to day.
- **Subject** — head sizes differ; `C3` is not the same point.
- **Montage** — 2 to 256 electrodes in different layouts.

Codex models (LaBraM, EEGPT): identity-indexed embeddings — **cannot even run** on an unseen layout.
Channel-independent (BIOT): no spatial structure at all.

Neither uses scalp geometry as an *input*.



The geometry has changed — per-electrode parameters canno

Solution: spatial attention conditioned on g_{ij}

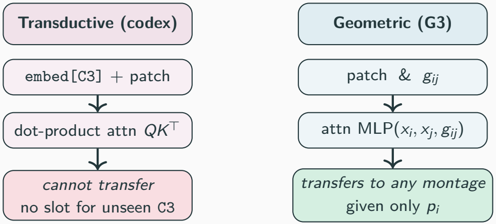
Geometric descriptor.

$$g_{ij} = \big[\, p_i,\ p_j,\ p_i - p_j,\ \|p_i - p_j\| \,\big] \in \mathbb{R}^{10}$$

Positions + displacement + distance. No per-electrode parameters; same shared MLP for *any* montage.

Injection fixed to G3 (geometry at *both* score and value – strongest in my earlier ablation). I instead ablate the descriptor: **pos_only** $= [p_i, p_j] \in \mathbb{R}^6$.

The only rival that can *also* transfer is **chind** (no geometry); the codex **cannot be evaluated** zero-shot.



raw EEG → patch → spatial attn → temporal → momentum target (EMA)

Experiment: leave-one-montage-out, zero-shot transfer

Train on a **mix of montages**; hold out one entire dataset; freeze the encoder; fit a linear probe on the **held-out montage** (genuinely zero-shot).

Identical training for every variant — only the spatial encoder differs (fair comparison).

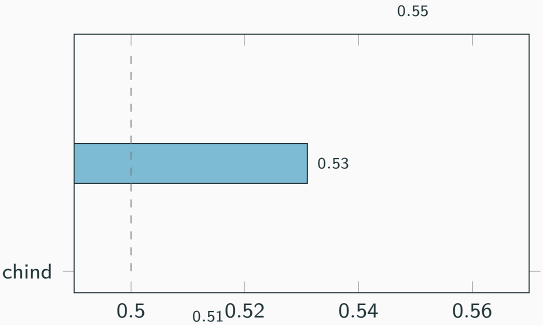
Balanced 9-subject-per-dataset corpus: geometric $\geq$ **chind** on **3/3** held-out montages. Direction is consistent; gaps are within noise at this scale.

Zero-shot BAC (balanced 9-subj corpus)

Held-out	geom	pos_only	chind
Sleep-EDFx	0.593	**0.597**	0.589
BCIC-2B	**0.509**	0.508	0.504
PhysioNet MI	0.303	**0.315**	0.302

Different tasks $\Rightarrow$ different chance levels; read *within* each row (geom vs chind), not across rows. Consistent but not yet significant — motivates scaling the one split we can.

Headline: at scale, geometric beats chind on zero-shot transfer



BCIC-2B zero-shot BAC (n=9 LOSO, n=156 pretrain)

Chance = 0.50 (binary). Dashed line = chance.

Scaled the one feasible split:
balanced n=156 (PhysioNet 78 + Sleep 78), zero-shot to BCIC-2B.

Geometric (full) = 0.545 beats chind (0.509): **+0.036**, $p = 0.023$, 7/9 subjects.

pos_only *also* beats chind: **+0.022**, $p = 0.013$, 8/9 subjects.

Codex is absent — it *cannot* be evaluated here at all. *That absence is the point.*

Conclusion: a clean positive, with clear edges

What I found:

- Geometry-conditioned attention **transfers zero-shot** to an unseen montage and **beats chind significantly** once given enough data.
- Robust to descriptor reduction (pos_only also significant).

Honest limits:

- Significant on **one** held-out montage (BCIC); small-corpus tests directional but underpowered (9-subject ceiling).

Future work: second distinct-montage held-out (e.g. 128 ch); larger corpus; implement the codex baseline.

Predicted vs. observed

	Predicted	**Obser**
Direction (3 montages)	geom +	**geom** +
Powered transfer	geom ++	**geom** +
Across all montages	hoped	1 powered;

`github.com/tianxin-scu/geometric-eeg-ssl`

Key References

- Wang et al. **EEGPT**. NeurIPS 2024 – training recipe (momentum + reconstruction).
- Grill et al. **BYOL**. NeurIPS 2020 – momentum-encoder alignment.
- He et al. **MAE**. CVPR 2022 – masked-patch reconstruction.
- Hamilton, Ying, Leskovec. **GraphSAGE**. NeurIPS 2017 – inductive vs. transductive vocabulary.
- Veličković et al. **GAT**. ICLR 2018 – attention over relational features.
- Jiang et al. **LaBraM**. ICLR 2024; Yang et al. **BIOT**. NeurIPS 2023 – codex / channel-independent baselines.
- Schalk et al. 2004 – PhysioNet MI; Tangermann et al. 2012 – BCIC-2B; Kemp et al. 2000 – Sleep-EDF.

Full reference list and per-experiment details in the project report.